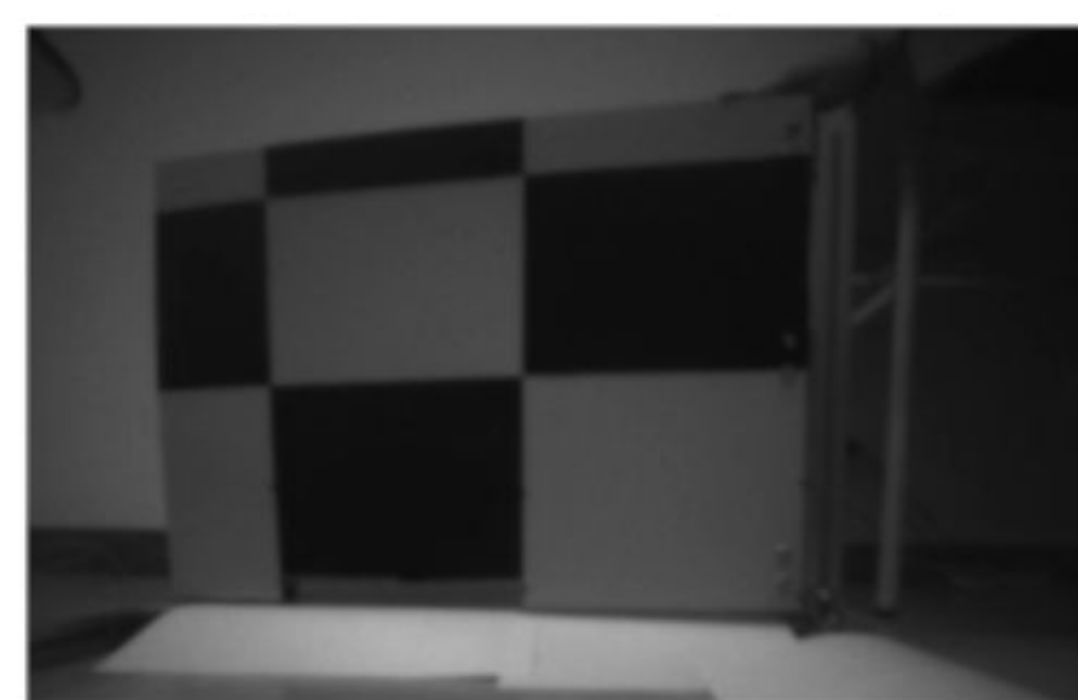
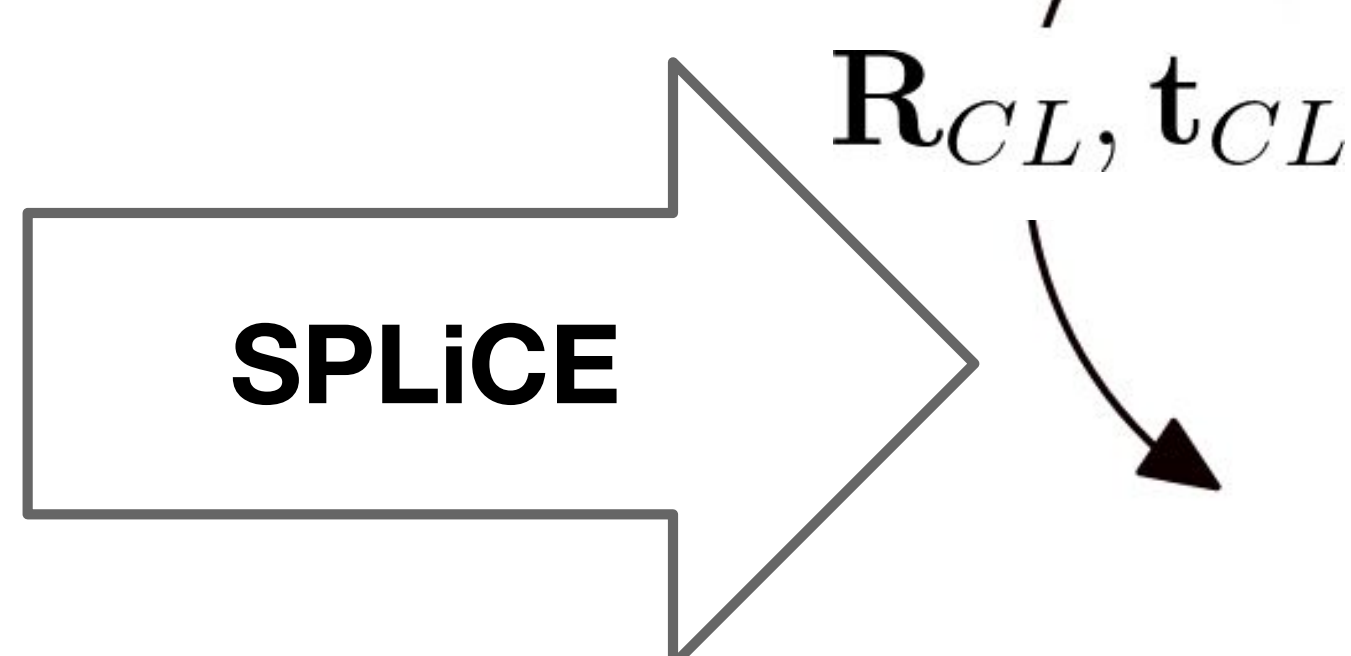


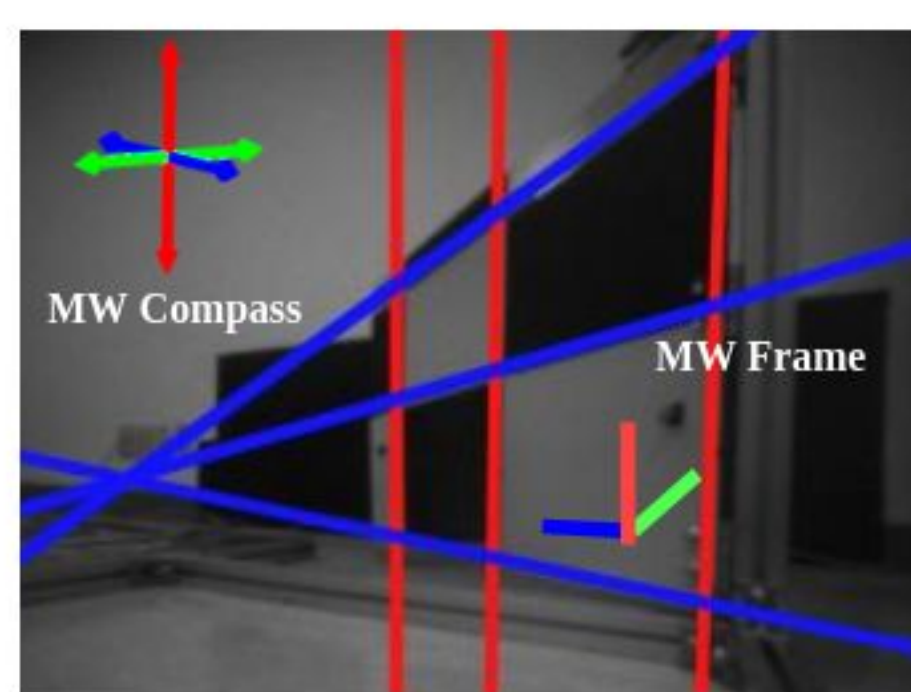
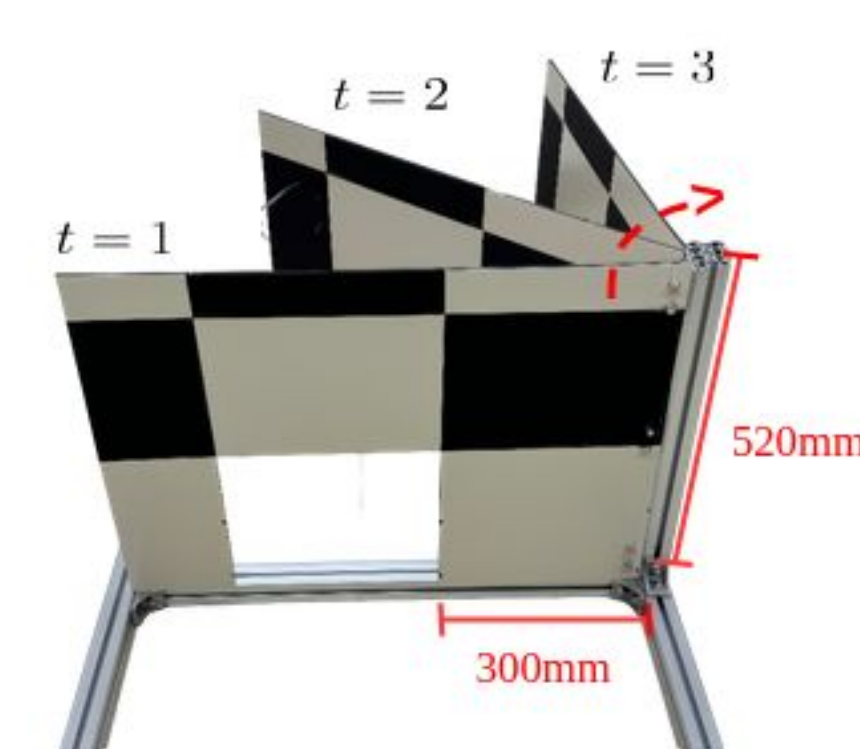
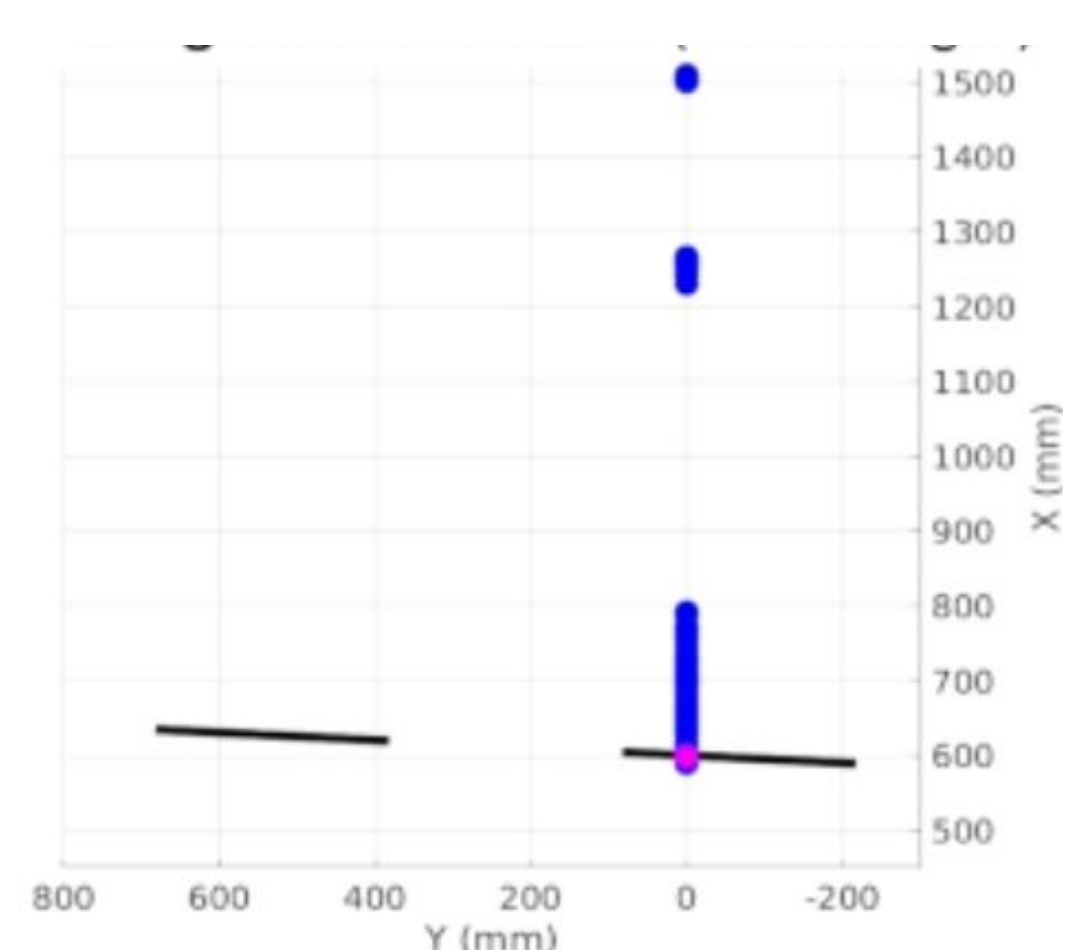
Motivation & Problem Setup

- Existing methods need **heavy, costly multi-beam LiDARs**
- Micro-robots** lack the resources to support such sensors
- Single-Point LiDAR** is lightweight but hard to calibrate

Image from Camera

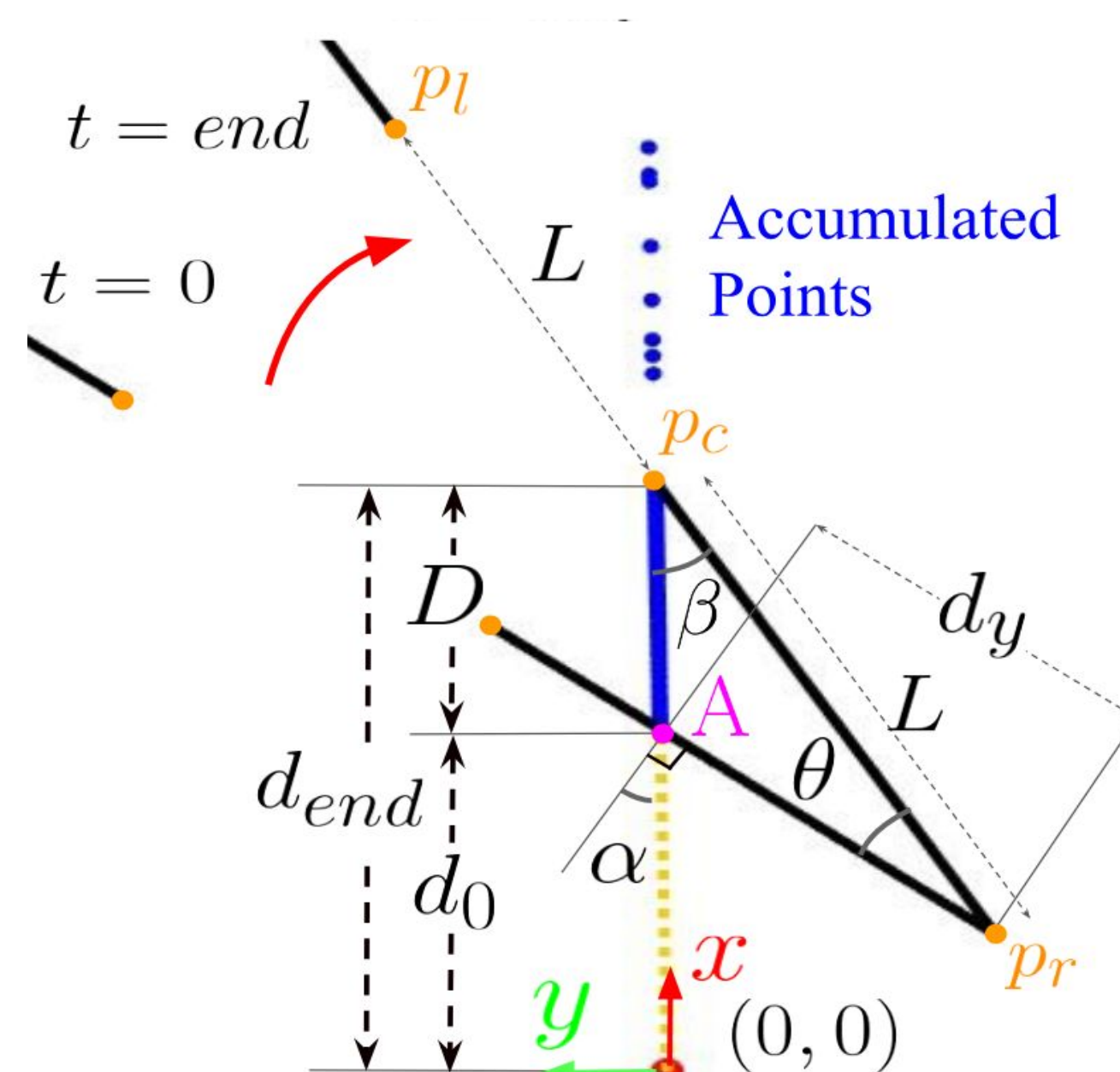
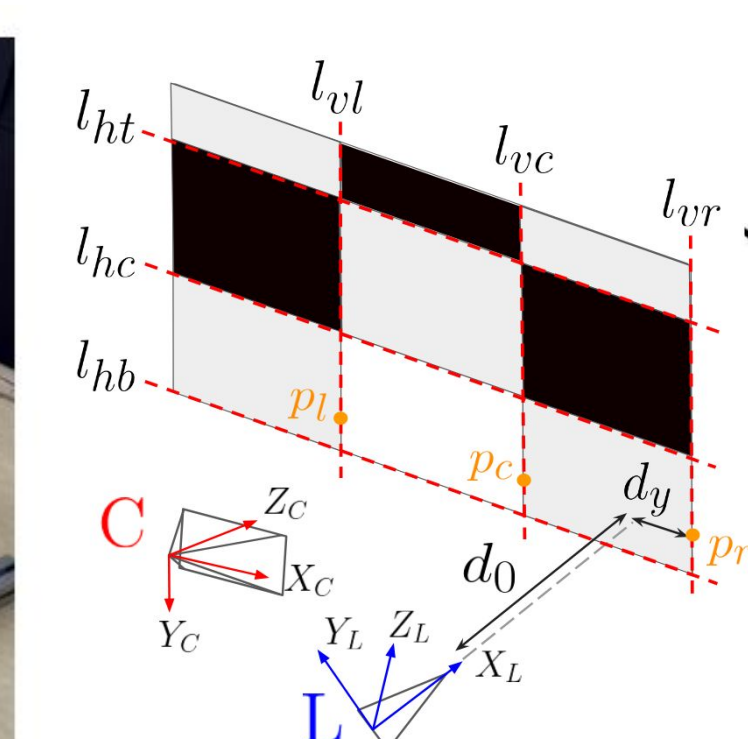
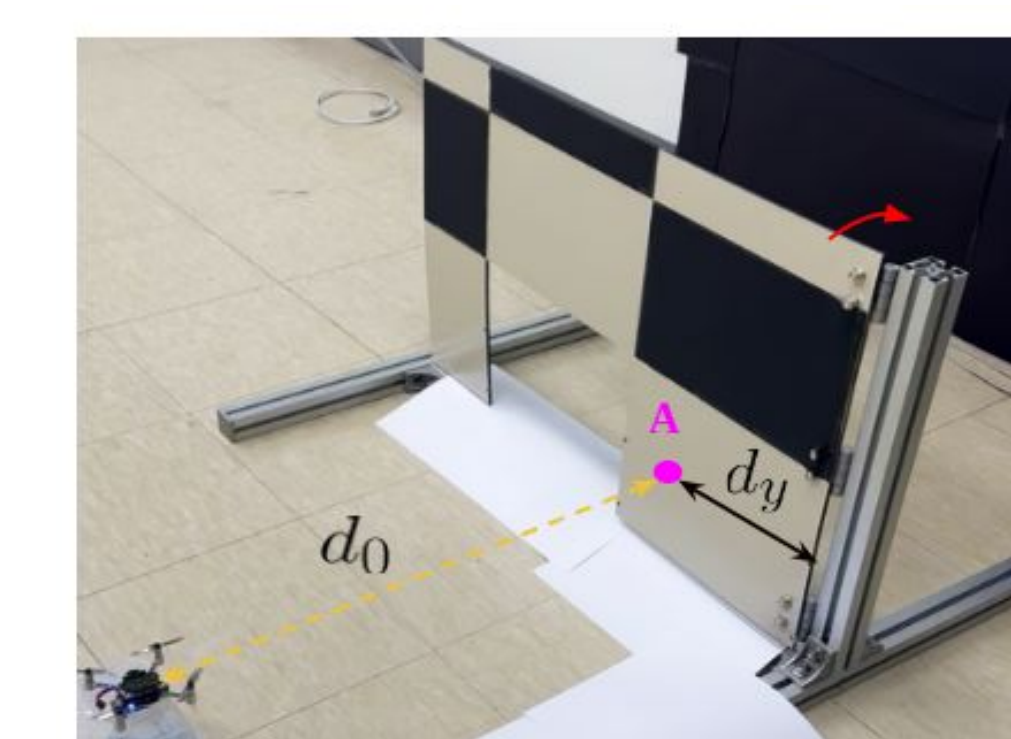

Single-Point LiDAR and Camera
Extrinsic
Parameters


- Custom calibration board satisfying the Manhattan World


Accumulated Points from
Single-Point LiDAR


Point-Line Constraints Optimization

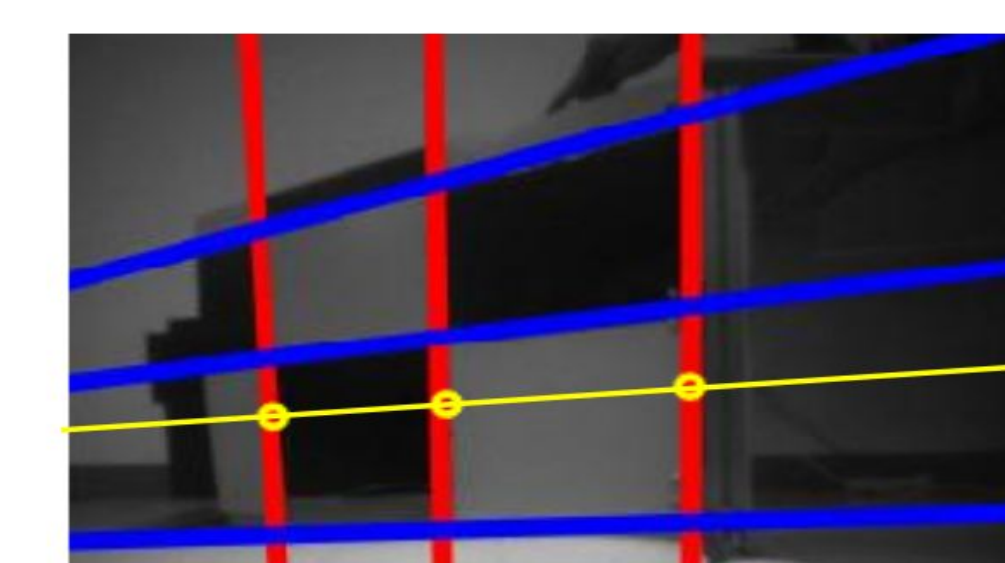
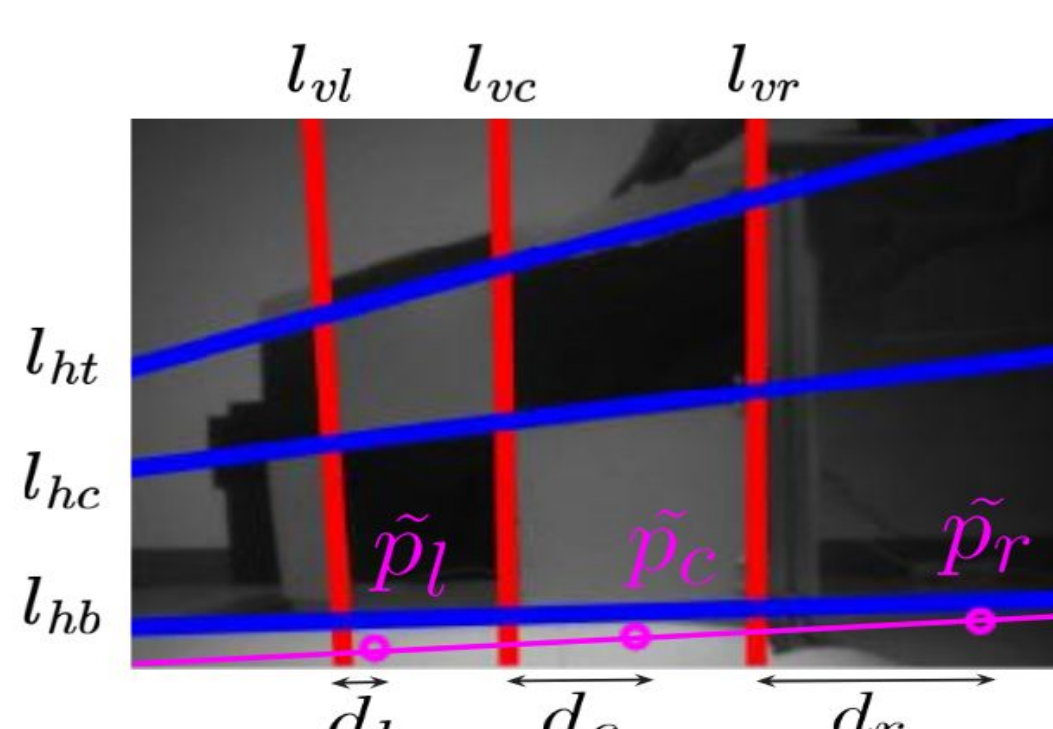
- We estimate **three key points** (p_r, p_c, p_l) by tracking LiDAR **distance** changes during board rotation and computing geometric parameters (θ, α, β)
- Minimizing **point-to-line distances** (d_k^i) between reprojected points ($\tilde{p}_l, \tilde{p}_c, \tilde{p}_r$) and detected vertical lines while aligning the **vanishing point** (p_j^i) with horizontal lines, optimized via the **LM** method.



$$\theta = \arcsin \frac{D \sin \beta}{\sqrt{L^2 + D^2 - 2LD \cos \beta}}$$

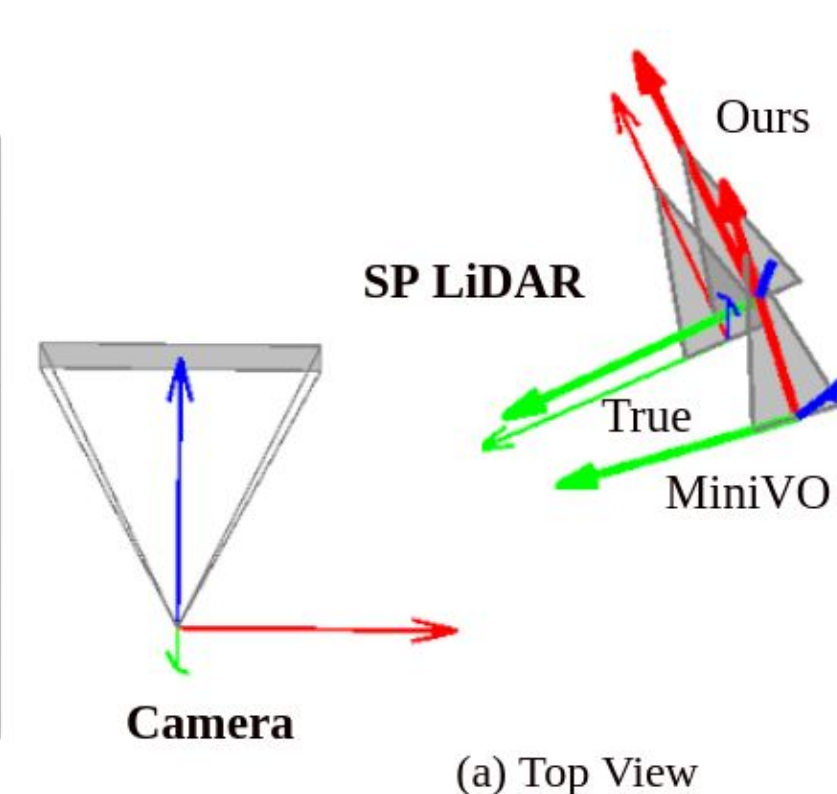
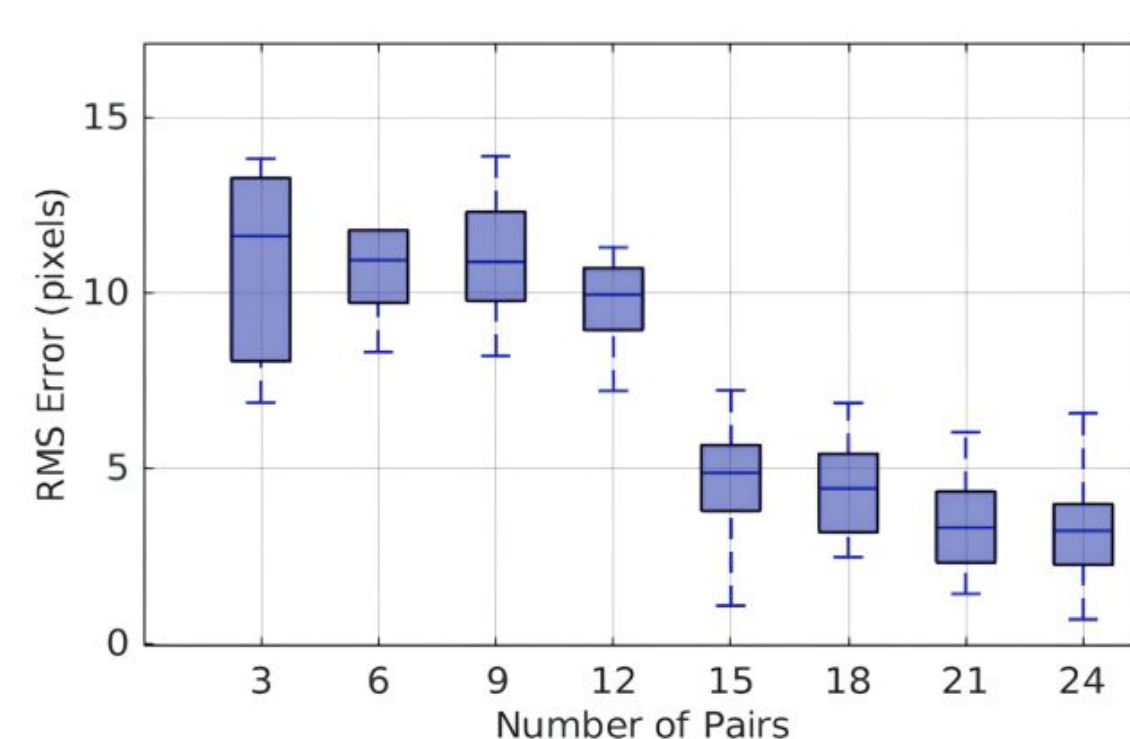
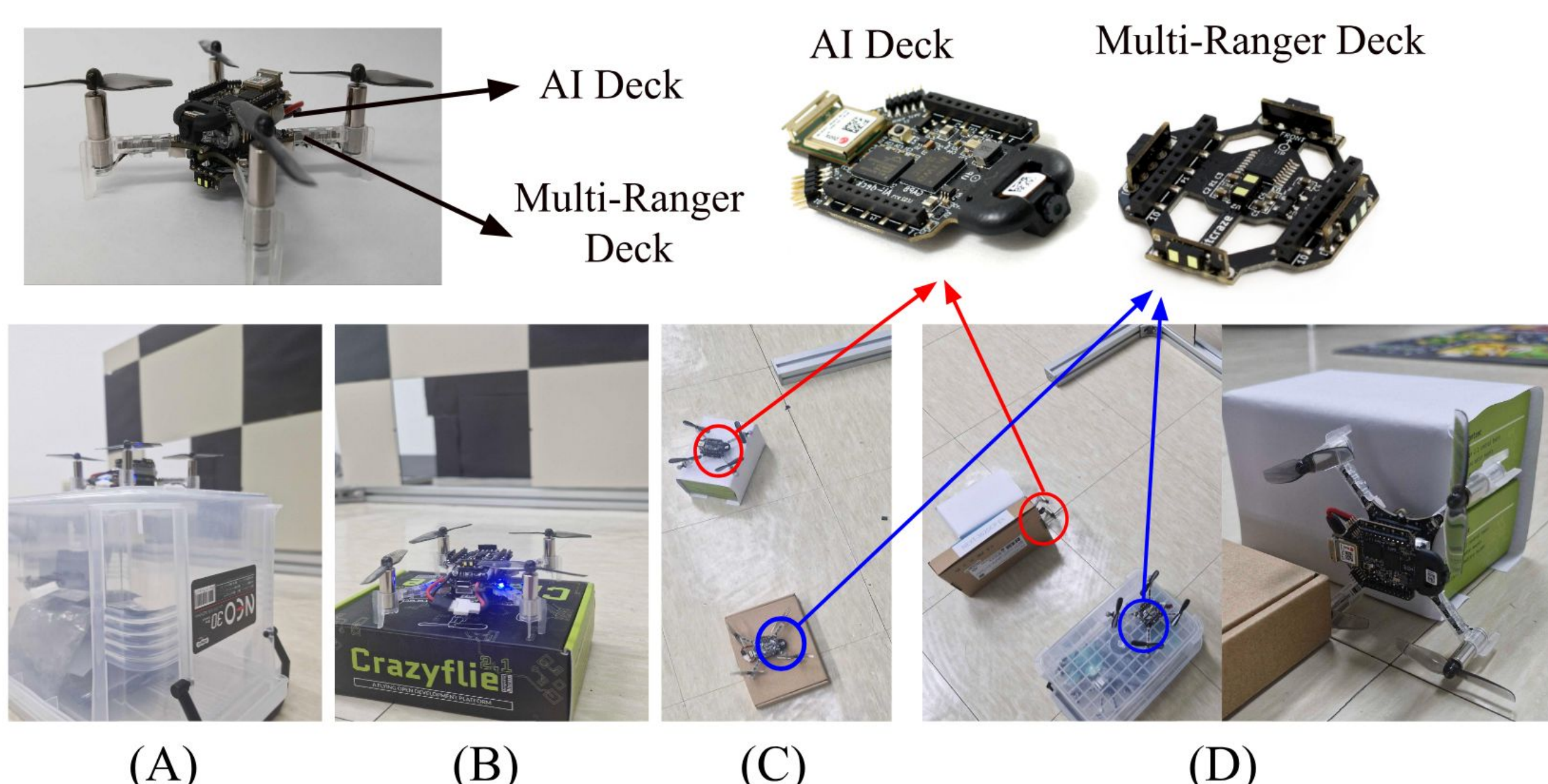
$$\mathbf{R}^*, \mathbf{t}^* = \arg \min_{\mathbf{R}, \mathbf{t}} \sum_{i=1}^N \left(\sum_{k=1}^3 (d_k^i)^2 + \sum_{j=1}^3 (p_c^i - p_j^i)^2 \right) \quad \mathbf{p}_c^i = \left(\frac{1}{3} \sum_{j=1}^3 u_j^i, \frac{1}{3} \sum_{j=1}^3 v_j^i \right)$$

(N: number of image-LiDAR pairs)

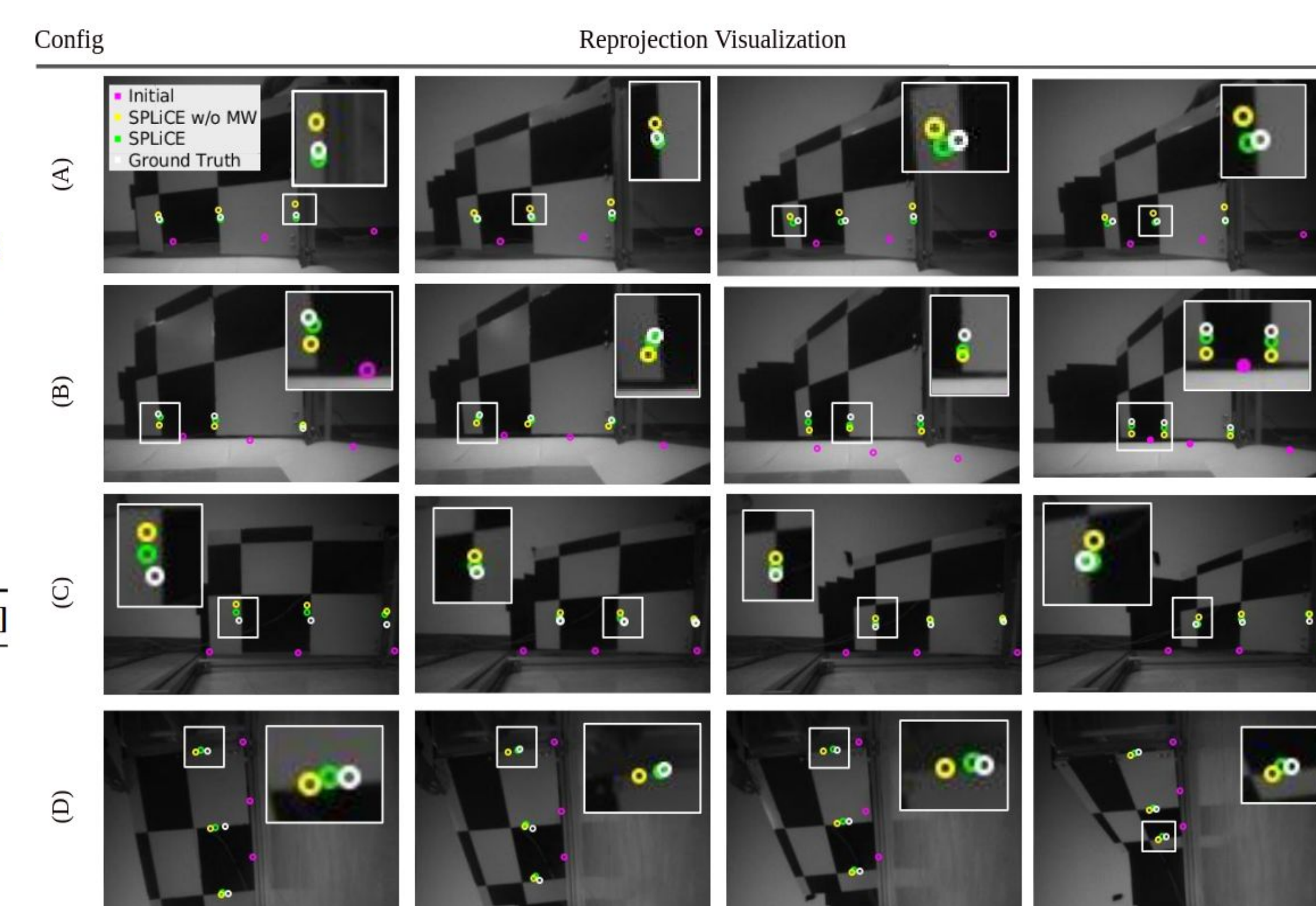


Evaluation

- Four Experimental Setups: **intra- and inter-drone**



Inter-Drone	tx [m]	ty [m]	tz [m]	roll [deg]	pitch [deg]	yaw [deg]
C						
SPLiCE w/o MW	0.0098	0.0110	0.0135	1.8344	1.9867	2.0475
SPLiCE	0.0098	0.0110	0.0135	1.1639	1.3869	1.2061
MiniVO	0.0253	0.0314	0.0288	-	6.3924	8.8846
D						
SPLiCE w/o MW	0.0149	0.0173	0.0187	2.0433	1.9001	2.7163
SPLiCE	0.0149	0.0173	0.0187	1.2314	1.0983	1.1863
MiniVO	0.0307	0.0696	0.0471	-	7.1024	9.4527



- The proposed SPLiCE achieves sub<3 pixel reprojection accuracy and sub<1° rotation error across all configurations
- These results confirm its effectiveness for reliable single-point LiDAR-camera calibration on lightweight micro-drone platforms